CHAPTER EXERCISE: Answers for this chapter start on page 285.

A calculator should NOT be used on the following questions.

1

A carpenter lays x bricks per hour for y hours and then lays $\frac{x}{2}$ bricks per hour for $2y$ more hours. In terms of x and y, how many bricks did he lay in total?

A) $2xy$

B) $\frac{5}{2}xy$

C) $5xy$

D) $\frac{3}{2}x + 3y$

2

A cheese vendor currently has 175 pounds of mozzarella available for sale. If each pound of mozzarella sells for \$8.75, which of the following functions gives the amount of mozzarella M, in pounds, still available for sale after d dollars worth has been sold?

A) $M(d) = 175 - \frac{d}{8.75}$

B) $M(d) = 175 - 8.75d$

C) $M(d) = 175 - \frac{8.75}{d}$

D) $M(d) = 175(8.75) - d$

3

A retail store has monthly fixed costs of \$3,000 and monthly salary costs of \$2,500 for each employee. If the store hires x employees for an entire year, which of the following equations represents the store's total cost c, in dollars, for the year?

A) $c = 3,000 + 2,500x$

B) $c = 12(3,000 + 2,500x)$

C) $c = 12(3,000) + 2,500x$

D) $c = 3,000 + 12(2,500x)$

4

An internet service provider charges a one time setup fee of \$100 and \$50 each month for service. If c customers join at the same time and are on the service for m months, which of the following expressions represents the total amount, in dollars, the provider has charged these customers?

A) $100c + 50m$

B) $100c + 50cm$

C) $150cm$

D) $100m + 50cm$

5

At a math team competition, there are m schools with n students from each school. The host school wants to order enough pizza such that there are 2 slices for each student. If there are 8 slices in one pizza, which of the following gives the number of pizzas the host school must order?

A) $\frac{mn}{8}$

B) $\frac{mn}{4}$

C) $\frac{m+2n}{8}$

D) $2mn$

6

A manufacturing plant increases the temperature of a chemical compound by d degrees Celsius every m minutes. If the compound has an initial temperature of t degrees Celsius, which of the following expressions gives its temperature after x minutes, in degrees Celsius?

A) $\frac{mx+t}{d}$

B) $\frac{md+t}{x}$

C) $t+\frac{d}{mx}$

D) $t+\frac{dx}{m}$

7

A cupcake store employs bakers to make boxes of cupcakes. Each box contains x cupcakes and each baker is expected to produce y cupcakes each day. Which of the following expressions gives the number of boxes needed for all the cupcakes produced by $3x$ bakers working for 4 days?

A) $12x^2y$

B) $\frac{3y}{4}$

C) $\frac{12x^2}{y}$

D) $12y$

8

At a shop for tourists, the price of one souvenir is a dollars. Each additional souvenir purchased after the first is discounted by 40 percent. Which of the following equations gives the total cost C, in dollars, of purchasing n souvenirs, where $n > 1$?

A) $C = a + (n-1)(0.4a)$

B) $C = a + (n-1)(0.6a)$

C) $C = a + n(0.6a)$

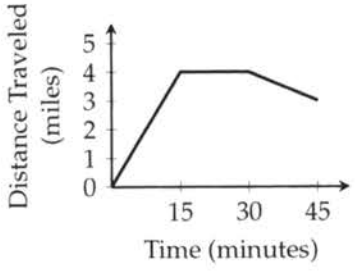
D) $C = 0.6an$

A calculator is allowed on the following questions.

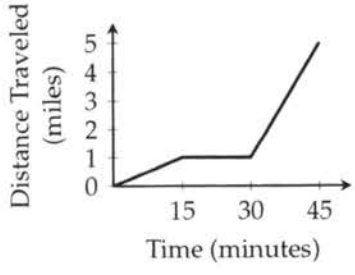
9

Kaiba began a 5-mile commute by biking for 4 miles to a rest area. He stopped at the rest area for 15 minutes and then walked for the remainder of the commute. If Kaiba bikes faster than he walks, which of the following graphs could represent his commute?

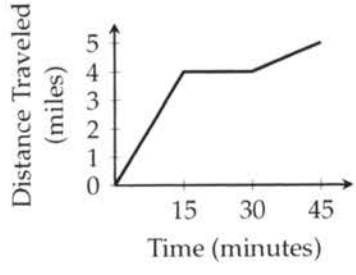
A)



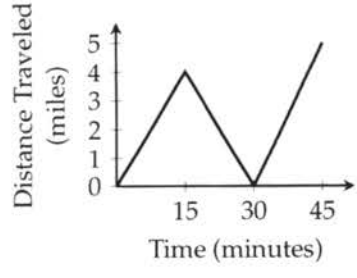
B)



C)



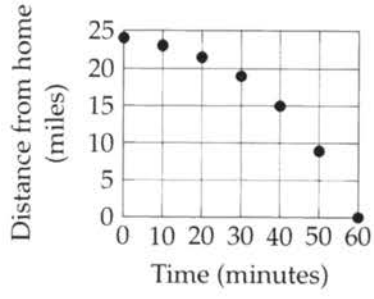
D)



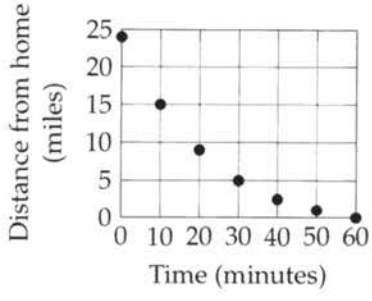
10

Mike starts driving to work and records his distance from home, in miles, every 10 minutes. His distance from home increases slowly at first due to traffic, then increases more quickly as traffic clears up. Which of the following graphs could illustrate Mike's distance from home during his drive?

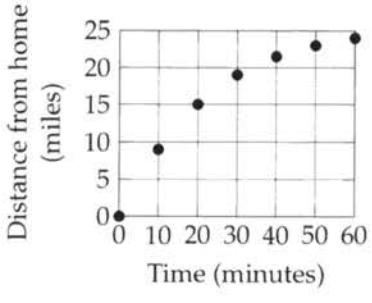
A)



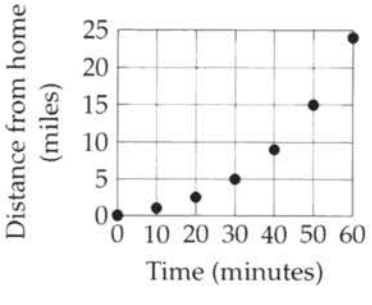
B)



C)



D)



11

At a video game arcade, d dollars can be exchanged for p tokens. If each game requires w tokens to play, which of the following gives the cost per game, in dollars?

A) $\frac{w}{dp}$

B) $\frac{d}{pw}$

C) $\frac{dw}{p}$

D) $\frac{dp}{w}$

12

To prepare for landing, a plane descends so that its altitude decreases at a constant rate from 24,500 feet to 17,900 feet in 12 minutes. Which of the following equations gives the altitude A, in feet, of the plane t minutes after its descent began, for $0 \leq t \leq 12$?

A) $A = 17{,}900 - 550t$

B) $A = 17{,}900 + 550t$

C) $A = 24{,}500 - 550t$

D) $A = 24{,}500 + 550t$

13

A taxicab charges a dollars for the first mile traveled and b dollars for each additional mile. If a particular passenger traveled more than one mile during a ride that cost \$24, which of the following represents the distance, in miles, the passenger traveled during the ride?

A) $\frac{24 - a - b}{b}$

B) $\frac{24 - a + b}{b}$

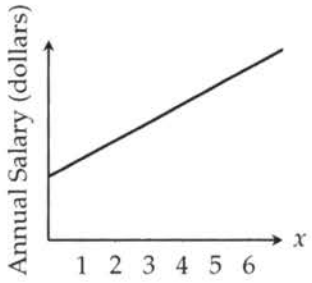
C) $\frac{24 + a - b}{b}$

D) $\frac{24 - a}{b}$

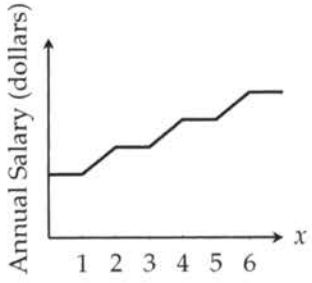
14

Mark started working as an inspector for a large construction company on June 1, 2019. According to his contract, his annual salary will increase by $15,000 on the first day of June each year. Which of the following graphs could model Mark's annual salary, in dollars, x years after June 1, 2019?

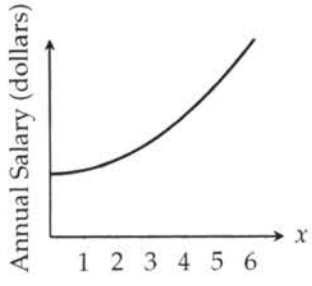
A)



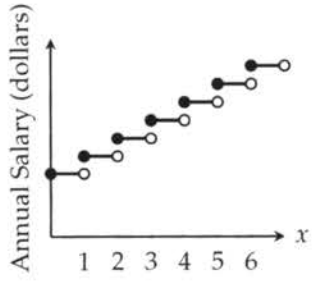
B)



C)



D)



15

To move into a new studio space, the m members of an art club agreed to split the first month's rent of r dollars equally among themselves. If k of the members fail to pay their share, which of the following represents the additional amount, in dollars, that each of the remaining members must pay to cover the first month's rent?

A) $\dfrac{r}{m-k}$

B) $\dfrac{kr}{m-k}$

C) $\dfrac{kr(m-k)}{m}$

D) $\dfrac{kr}{m(m-k)}$

Chapter 7: Constructing Models

CHAPTER EXERCISE:

1. [A] In the first y hours, the carpenter lays $(x)(y) = xy$ bricks. In the $2y$ hours thereafter, he lays $\left(\frac{x}{2}\right)(2y) = xy$ bricks. Altogether, that's $xy + xy = 2xy$ bricks.

2. [A] For d dollars worth of mozzarella to have been sold, $\frac{d}{8.75}$ pounds must have been sold. That leaves $175 - \frac{d}{8.75}$ pounds still available for sale.

3. [B] The store's monthly total cost is $3,000 + 2,500x$. For an entire year, we multiply by 12 months: $c = 12(3,000 + 2,500x)$.

4. [B] The setup fees amount to $100c$, \$100 for each customer. The monthly cost for all the customers amounts to $50c$, \$50 for each customer. Over m months, the monthly charges add up to $50c \times m$, or $50cm$. The total charge is therefore $100c + 50cm$.

5. [B] For mn students, the total number of slices must be $2mn$. Since there are 8 slices in each pizza, the school must order $\frac{2mn}{8} = \frac{mn}{4}$ pizzas.

6. [D] The compound's temperature increases by $\frac{d}{m}$ degrees per minute. So after x minutes, the temperature increases by $\frac{d}{m}x$, or $\frac{dx}{m}$. The final temperature is then $t + \frac{dx}{m}$.

7. [D] The bakers make $3xy$ cupcakes each day. Over 4 days, they will make a total of $4 \times 3xy = 12xy$ cupcakes. The number of boxes needed is the total number of cupcakes divided by the number of cupcakes that can fit in each box: $\frac{12xy}{x} = 12y$.

8. [B] The reduced price of each souvenir after the first is $0.6a$. So the first souvenir costs a dollars and the remaining $n - 1$ souvenirs each cost $0.6a$ dollars. Therefore, the total cost is $a + (n - 1)(0.6a)$.

9. [C] During the biking portion of the commute, the graph should go up and be relatively steep since Kaiba covers the initial 4 miles at a faster pace than he walks. When Kaiba stops at the rest area for 15 minutes, the graph should be flat since he does not cover any distance during this time. After Kaiba leaves the rest area, the graph should go up and be at a gradual incline since he covers the last mile of his commute at a walking pace. Only the graph in answer C fulfills all of the above criteria.

10. [D] Since Mike's distance from home increases during his commute, we're looking for a graph that goes up and to the right. Since his distance from home increases slowly at first and then more quickly later, we're also looking for the graph to go from a low slope (less steep) to a high slope (more steep). Only the graph in answer D meets these conditions.

11. [C] Since p tokens can be used to play $\frac{p}{w}$ games, the cost per game is $\frac{d}{\frac{p}{w}} = \frac{dw}{p}$ dollars.

12. [C] The plane descends at a rate of $\frac{24,500 - 17,900}{12} = \frac{6,600}{12} = 550$ feet per minute. Since the plane started its descent at an altitude of 24,500 feet, its altitude after t minutes can be represented by $A = 24,500 - 550t$.

13. [B] The passenger spent $24 - a$ dollars on additional miles after the first, which means the passenger traveled $\frac{24 - a}{b}$ additional miles after the first. Adding the first mile then gives us the total distance traveled:

$$\frac{24 - a}{b} + 1 = \frac{24 - a}{b} + \frac{b}{b} = \frac{24 - a + b}{b}$$

14. [D] Mark's annual salary does not change during the year except for June 1st, when it "jumps" by \$15,000. Therefore, the correct graph should be flat throughout each year except for a jump at the end of each year (since we are starting from June 1st, the end of each year refers to the next June 1st). Only the graph in answer D fits this description.

15. [D] Initially, the members are each responsible for paying $\frac{r}{m}$ dollars. But if k members fail to pay, then $m - k$ members remain and each one becomes responsible for $\frac{r}{m - k}$ dollars. This amount is greater than the original amount by

$$\frac{r}{m-k} - \frac{r}{m} = \frac{mr}{m(m-k)} - \frac{r(m-k)}{m(m-k)} = \frac{mr - mr + kr}{m(m-k)} = \frac{kr}{m(m-k)} \text{ dollars}$$

Here is an alternative solution. Since each member is initially responsible for $\frac{r}{m}$ dollars, the club loses out on $\frac{r}{m}(k)$ dollars when the k members fail to pay. To make up this lost amount, each of the remaining $m - k$ members must pay an additional

$$\frac{\frac{r}{m}(k)}{m-k} = \frac{\frac{kr}{m}}{m-k} = \frac{kr}{m(m-k)} \text{ dollars}$$